

Tonmya_Prescriptions

2026-04-12

Some simple models based on Tonmya sales

What are we doing here? - Just trying to get a grasp on the rate of increase and projecting out if that rate continues. It is not necessarily realistic to take it out as far as projections provide. Nothing here should be regarded as investment advice. I am not a financial analyst or advisor.

Sales Projection - at weekly growth rate - *exponential model*

Formula: $\text{sales} \sim a * (1 + r)^{\text{wk}}$

Parameters:

	Estimate	Std. Error	t value	Pr(> t)
a	98.64503	15.36756	6.419	4.83e-06 ***
r	0.12103	0.01042	11.616	8.50e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

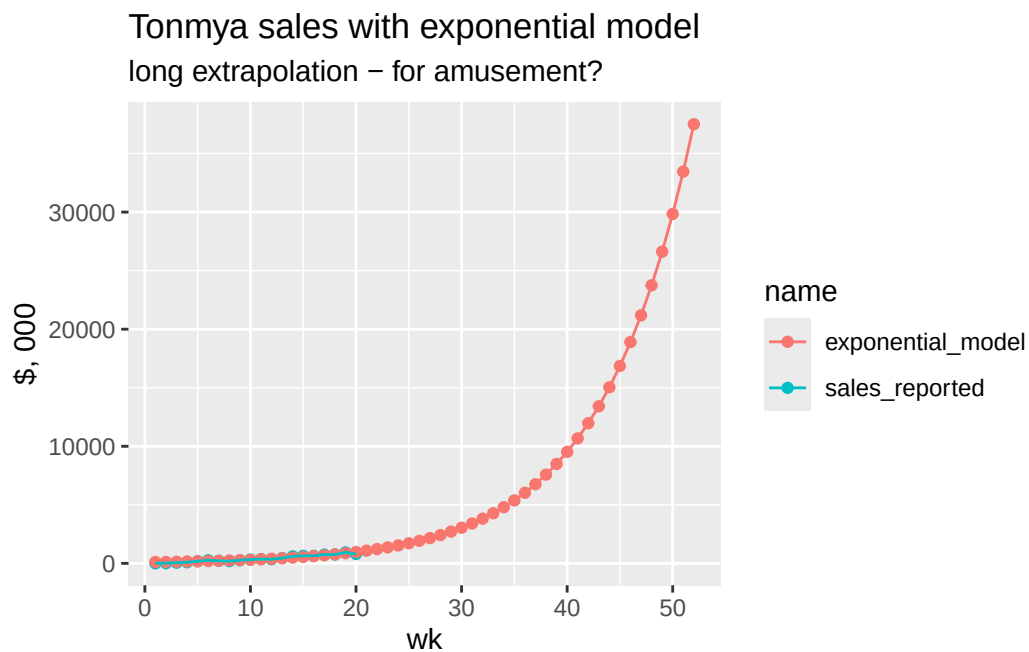
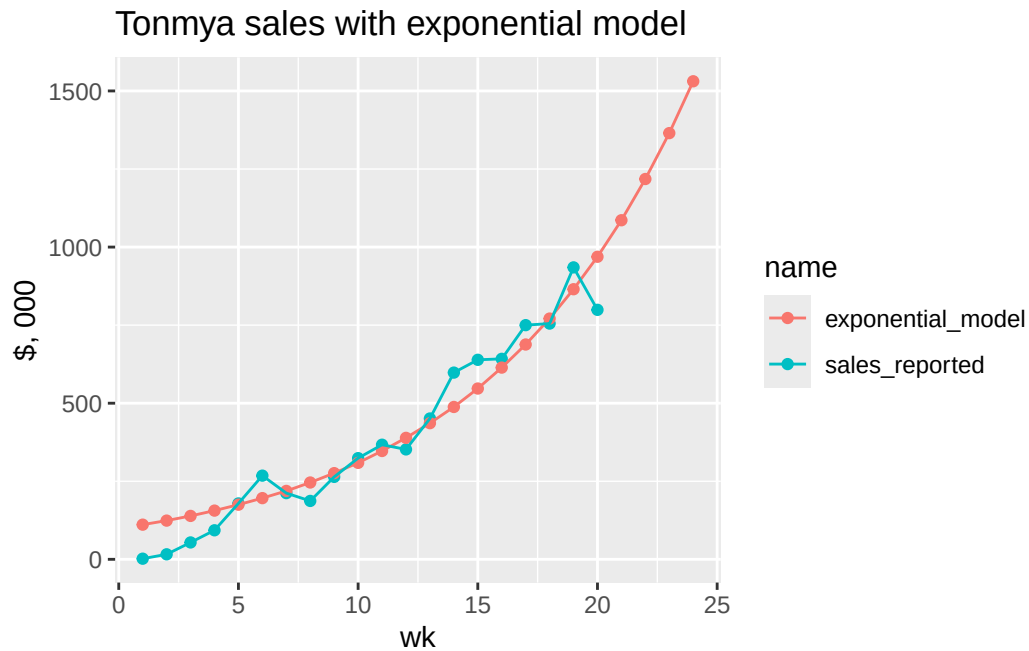
Residual standard error: 76.37 on 18 degrees of freedom

Number of iterations to convergence: 16

Achieved convergence tolerance: 3.166e-06

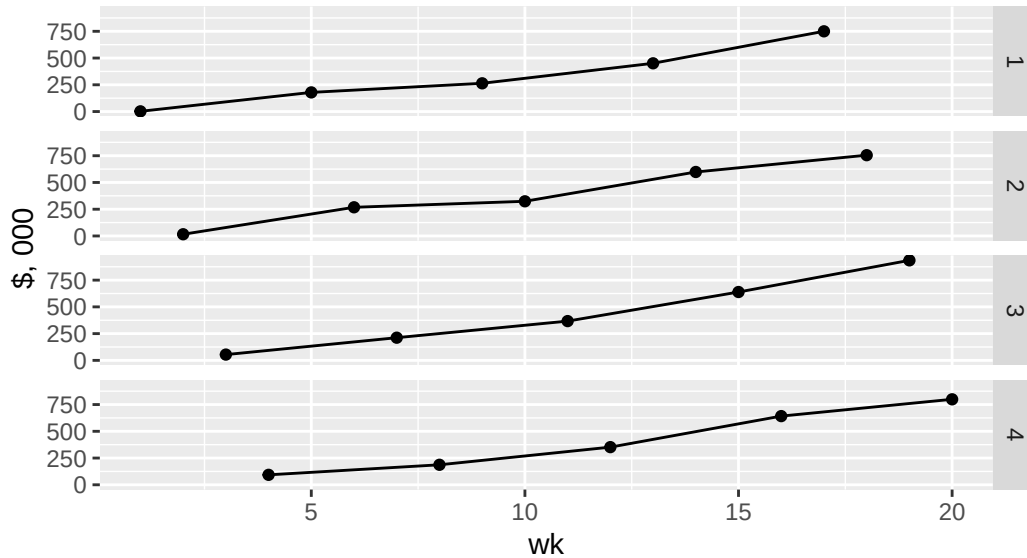
Exponential Model

wk	sales, 000	exp, 000	annual, M
1	\$2	\$111	\$5.8
2	\$16	\$124	\$6.4
3	\$54	\$139	\$7.2
4	\$93	\$156	\$8.1
5	\$179	\$175	\$9.1
6	\$268	\$196	\$10.2
7	\$212	\$219	\$11.4
8	\$187	\$246	\$12.8
9	\$264	\$276	\$14.3
10	\$324	\$309	\$16.1
11	\$367	\$347	\$18.0
12	\$352	\$389	\$20.2
13	\$451	\$436	\$22.7
14	\$598	\$488	\$25.4
15	\$639	\$547	\$28.5
16	\$642	\$614	\$31.9
17	\$750	\$688	\$35.8
18	\$755	\$771	\$40.1
19	\$935	\$865	\$45.0
20	\$799	\$969	\$50.4
21	NA	\$1,086	\$56.5
22	NA	\$1,218	\$63.3
23	NA	\$1,365	\$71.0
24	NA	\$1,531	\$79.6



I think we can see some trending on a monthly basis. Reporting doesn't match a monthly basis, but we can see if four-week intervals are interesting.

Tonmya sales with exponential model broken into week cohorts 1 through 4



Sales Projection - at weekly rate - *linear model*

Call:

```
lm(formula = sales ~ wk, data = df_sales)
```

Residuals:

Min	1Q	Median	3Q	Max
-112.932	-47.836	6.123	40.081	140.683

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-99.726	29.553	-3.374	0.00338 **
wk	47.055	2.467	19.073	2.18e-13 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

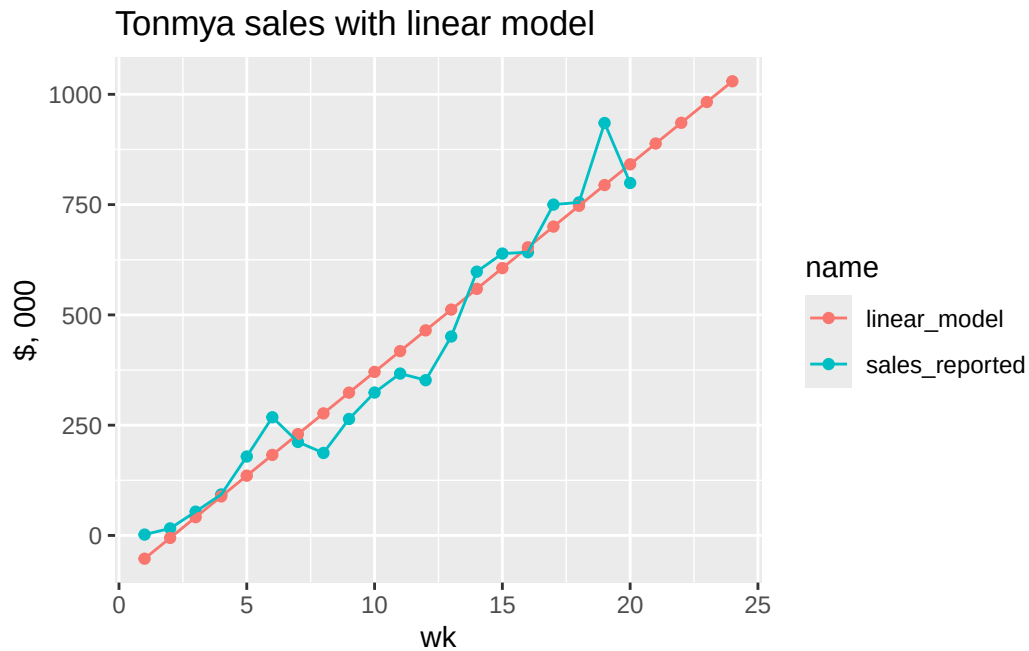
Residual standard error: 63.62 on 18 degrees of freedom

Multiple R-squared: 0.9529, Adjusted R-squared: 0.9502

F-statistic: 363.8 on 1 and 18 DF, p-value: 2.181e-13

Linear Model

wk	sales, 000	exp, 000	annual, M
1	\$2	-\$53	\$2.4
2	\$16	-\$6	\$4.9
3	\$54	\$41	\$7.3
4	\$93	\$88	\$9.8
5	\$179	\$136	\$12.2
6	\$268	\$183	\$14.7
7	\$212	\$230	\$17.1
8	\$187	\$277	\$19.6
9	\$264	\$324	\$22.0
10	\$324	\$371	\$24.5
11	\$367	\$418	\$26.9
12	\$352	\$465	\$29.4
13	\$451	\$512	\$31.8
14	\$598	\$559	\$34.3
15	\$639	\$606	\$36.7
16	\$642	\$653	\$39.1
17	\$750	\$700	\$41.6
18	\$755	\$747	\$44.0
19	\$935	\$794	\$46.5
20	\$799	\$841	\$48.9
21	NA	\$888	\$51.4
22	NA	\$935	\$53.8
23	NA	\$983	\$56.3
24	NA	\$1,030	\$58.7



Scripts Projection - at weekly growth rate - *exponential model*

Formula: $\text{scripts} \sim a * (1 + r)^{\text{wk}}$

Parameters:

	Estimate	Std. Error	t value	Pr(> t)
a	58.029083	8.278472	7.01	1.53e-06 ***
r	0.125041	0.009527	13.12	1.18e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

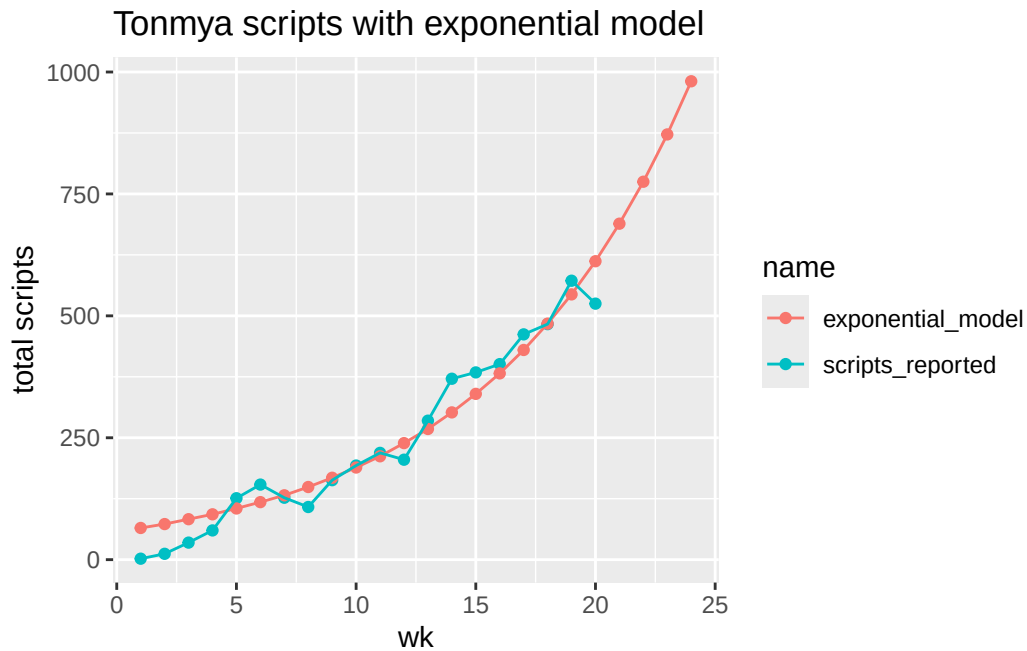
Residual standard error: 42.56 on 18 degrees of freedom

Number of iterations to convergence: 10

Achieved convergence tolerance: 2.533e-06

Scripts, Exponential Model

wk	total reported, 000	exp predicted, 000	annual, M
1	2	65	3.4
2	12	73	3.8
3	35	83	4.3
4	60	93	4.8
5	126	105	5.4
6	154	118	6.1
7	127	132	6.9
8	108	149	7.7
9	163	168	8.7
10	193	189	9.8
11	219	212	11.0
12	205	239	12.4
13	285	268	14.0
14	371	302	15.7
15	384	340	17.7
16	401	382	19.9
17	462	430	22.4
18	483	484	25.2
19	572	544	28.3
20	525	612	31.8
21	NA	689	35.8
22	NA	775	40.3
23	NA	872	45.3
24	NA	981	51.0



scripts Projection - at weekly rate - *linear model*

Call:

```
lm(formula = scripts ~ wk, data = df_scripts)
```

Residuals:

Min	1Q	Median	3Q	Max
-83.609	-33.986	7.156	23.999	76.849

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-65.463	18.881	-3.467	0.00275 **
wk	29.506	1.576	18.720	3.01e-13 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40.64 on 18 degrees of freedom

Multiple R-squared: 0.9511, Adjusted R-squared: 0.9484

F-statistic: 350.5 on 1 and 18 DF, p-value: 3.007e-13

Scripts, Linear Model

wk	total reported, 000	linear predicted, 000	annual, M
1	2	-36.0	1.5
2	12	-6.5	3.1
3	35	23.1	4.6
4	60	52.6	6.1
5	126	82.1	7.7
6	154	111.6	9.2
7	127	141.1	10.7
8	108	170.6	12.3
9	163	200.1	13.8
10	193	229.6	15.3
11	219	259.1	16.9
12	205	288.6	18.4
13	285	318.1	19.9
14	371	347.6	21.5
15	384	377.1	23.0
16	401	406.6	24.5
17	462	436.1	26.1
18	483	465.6	27.6
19	572	495.2	29.2
20	525	524.7	30.7
21	NA	554.2	32.2
22	NA	583.7	33.8
23	NA	613.2	35.3
24	NA	642.7	36.8

